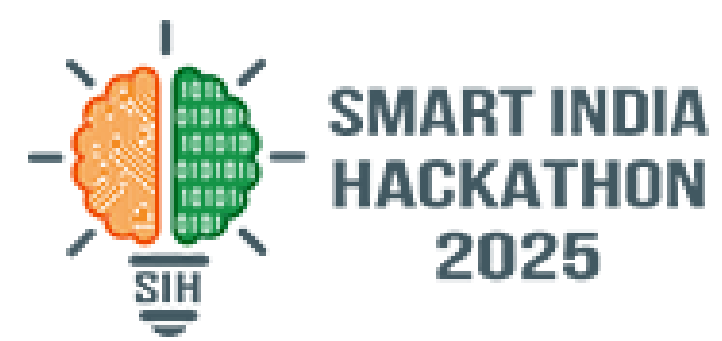


SMART INDIA HACKATHON 2025



- Problem Statement ID – 25101
- Problem Statement Title- Remote classroom for rural colleges
- Theme- Smart Education
- PS Category- Software
- Team ID-
- Team Name:- Goodfellas



We are developing a real-time classroom recording and streaming platform for remote education.

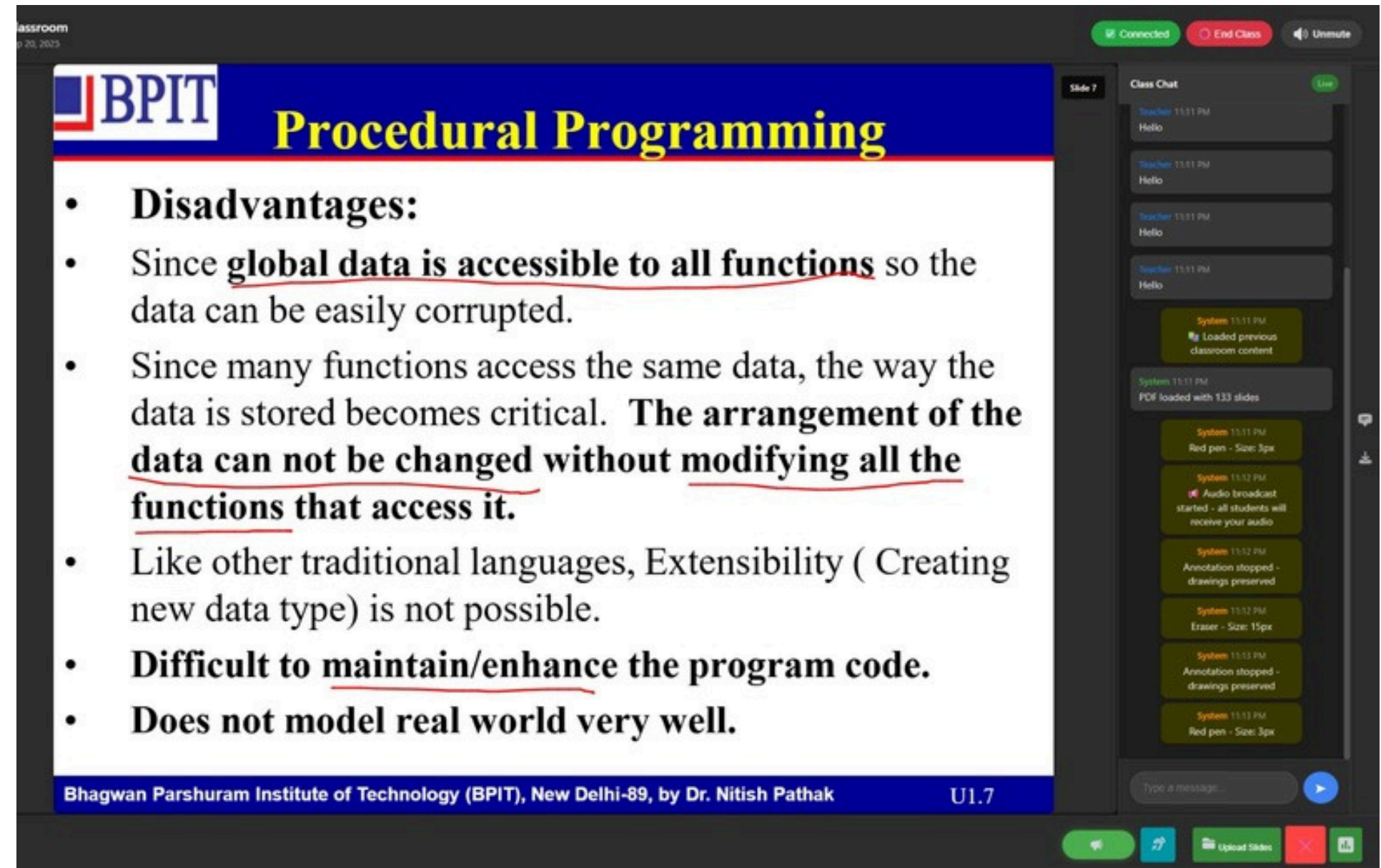
This solution will **capture teacher screens and audio** using **browser APIs running in real-time**. **Clean recordings** are processed and stored, while **students access lectures** through an intuitive dashboard with **detailed playback options** and **AI-based quizzes**.

 By combining screen + audio capture, we ensure comprehensive lecture recording, covering both visual presentations and teacher narration.

 It automatically records, processes, and stores lecture content, generating downloadable videos for students.

 The solution operates fully web-based, and is compatible with all browsers and devices including mobile phones and consumes ~150mb/hr

 A dashboard allows students to access previous lectures and download content on-demand.



classroom

10/20/2023

Connected End Class Unmute

Slide 7

Class Chat

Teacher 11:11 PM
Hello

Teacher 11:11 PM
Hello

Teacher 11:11 PM
Hello

Teacher 11:11 PM
Hello

System 11:11 PM
Loaded previous classroom content

System 11:11 PM
PDF loaded with 133 slides

System 11:11 PM
Red pen - Size: 3px

System 11:12 PM
Audio broadcast started - all students will receive your audio

System 11:12 PM
Annotation stopped - drawings preserved

System 11:12 PM
Eraser - Size: 15px

System 11:13 PM
Annotation stopped - drawings preserved

System 11:13 PM
Red pen - Size: 3px

Type a message

BPIT

Procedural Programming

- **Disadvantages:**
- Since global data is accessible to all functions so the data can be easily corrupted.
- Since many functions access the same data, the way the data is stored becomes critical. The arrangement of the data can not be changed without modifying all the functions that access it.
- Like other traditional languages, Extensibility (Creating new data type) is not possible.
- Difficult to maintain/enhance the program code.
- Does not model real world very well.

Bhagwan Parshuram Institute of Technology (BPIT), New Delhi-89, by Dr. Nitish Pathak

U1.7

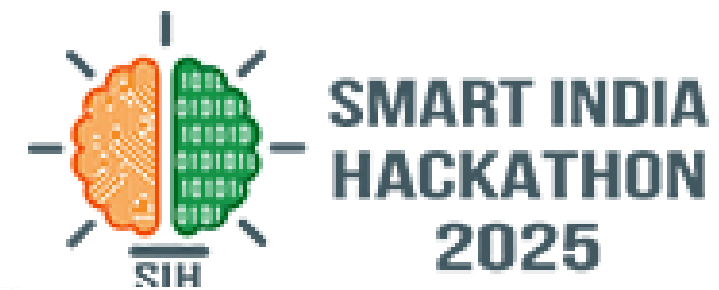
Upload Slides

Goodfellas

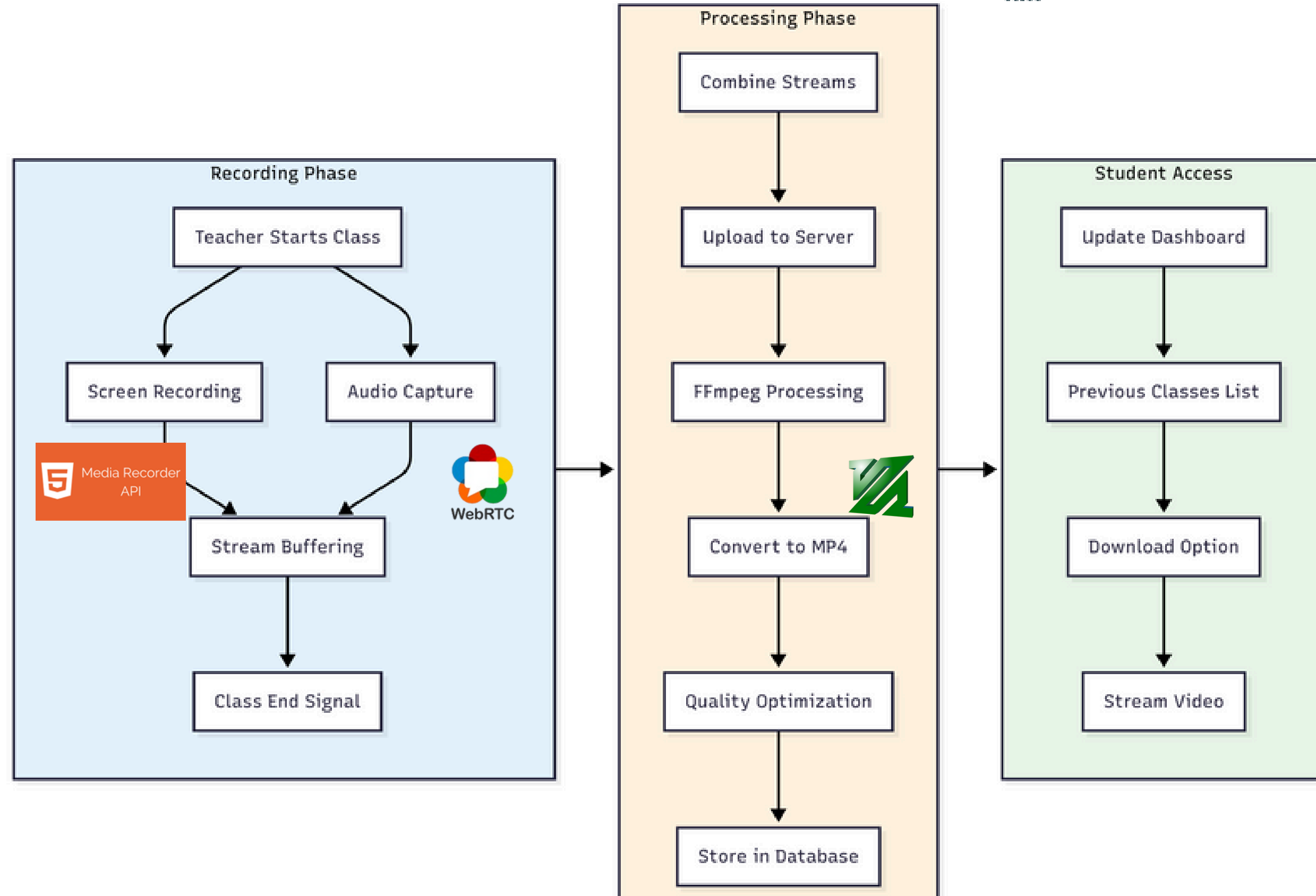


A Docker image of the AV service helps it run efficiently, and agnostic of system specifications.

TECHNICAL APPROACH



Frontend: HTML5/CSS3 (Material-UI), JavaScript (WebRTC Integration)
Backend: Go (Gorilla Mux/WebSocket Server)
Database: SQLite (Session & Media Storage)
Libraries: WebRTC, PDF.js, Canvas API, FFmpeg, GORM, bcrypt
Deployment: Docker Compose
Other: WebSocket, MediaRecorder API, Screen Capture API, STUN/TURN



Challenges

- 
- **Low bandwidth** connectivity limits clear audio/video transmission in rural areas, causing **frequent disconnections** and **poor learning experience**.
 - **Device limitations** vary by student type, with entry-level smartphones having **processing constraints** and **limited storage** capacity.
 - Teacher **digital literacy** presents adoption barriers as rural educators often **lack experience** with virtual platforms and require extensive training support.
 - **Infrastructure dependencies** create concerns due to unreliable internet and **slow speeds** and **overall low bandwidth availability**.

Solutions

- 
- Leverage **adaptive WebRTC streaming** by implementing **Opus codec** (6-40 kbps) with **dynamic bitrate adjustment** and audio-first fallback to ensure reliable connectivity.
 - Enhance platform reliability by normalizing **offline-first architecture** with **downloadable content**, **session recording**, and **Progressive Web App capabilities** for low-spec devices.
 - Expand affordable access through **device agnostic approach** and **widespread compatibility**.
 - **Low internet requirements** due to the audio first approach (~150Mb/Hr)

Technical Feasibility

- **Real-time Communication:** WebRTC with Opus codec provides efficient **8-48 kbps audio streaming**, with websockets for low latency chats and annotations.
- **Content Delivery:** Slide rendering using **Pdf.js** and **realtime annotations** from **Canvas API**.
- **Scalability:** Go backend with **Gorilla Mux** handles **concurrent connections** efficiently, session-based authentication ensures security, and **Docker containerization** enables easy deployment.

Economic Viability

- **Cost-Effective Model:** **Software-only solution** eliminates hardware requirements, **open-source technologies** eliminate licensing costs, and **cloud deployment** scales based on usage.
- **Localized Implementation:** Rural colleges can adopt the platform through **minimal infrastructure investment** locally or can **scale depending on usage** using cloud services such as AWS.

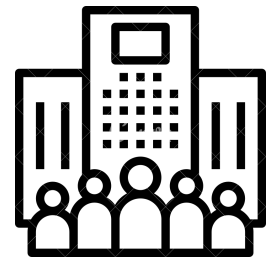
Stakeholder Aspect

- **Educators:** Need **reliable, cost effective and low-bandwidth solutions** which prioritize their needs and have low connection drop rates.
- **Rural Students:** Already familiar with smartphones and require a solution which allows them to **study with minimal internet usage and offline access** to study at their pace.
- **Educational Institutions:** Need a **low cost solution** which does not increase their expenditure by a considerable margin while ensuring that students get **access to quality education without compromises**.

Operational Viability

- **Self-Hosted Deployment:** Go binary + SQLite runs on **basic Linux server (2GB RAM, 20GB storage)**, eliminating cloud costs for institutions with existing hardware such as old desktop computers.
- **Docker Flexibility:** Deploy locally via docker or migrate seamlessly to cloud services if needed which auto scale.
- **Resource Efficiency:** Single <100MB binary handles WebSocket, WebRTC, and HTTP connections, enabling repurposed desktop computers as classroom servers

IMPACT ON TARGET AUDIENCE



Rural Colleges:

Enhanced access to specialized subject expertise, eliminating geographical barriers to quality educators.

Urban Faculty:

Opportunity to reach and impact underserved rural communities, expanding teaching reach beyond physical constraints while maintaining simple delivery methods

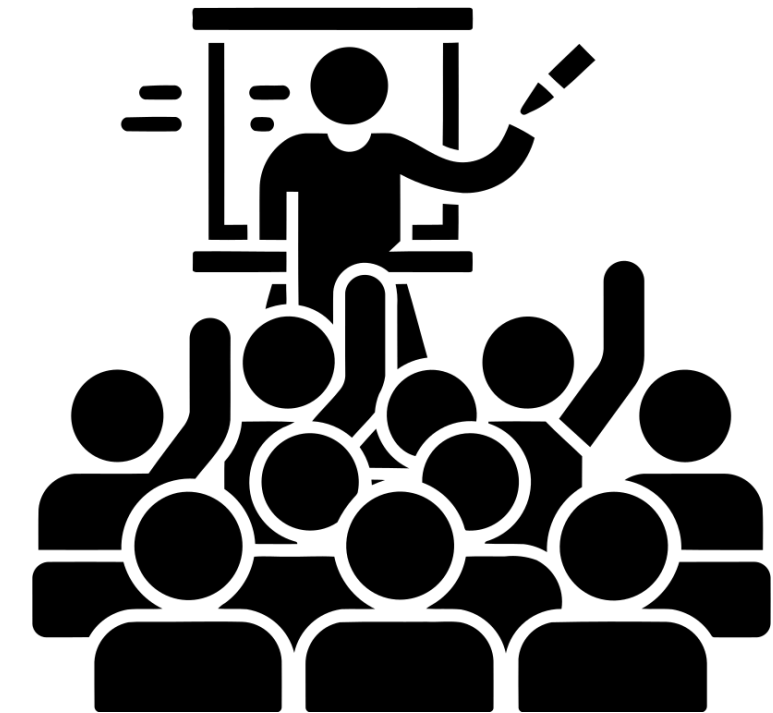
Rural Students:

Improved academic outcomes through interactive virtual classrooms, real-time collaboration with teachers from technical backgrounds, and flexible learning schedules that accommodate their time.



BENEFITS

- Bridges the **urban-rural education gap** by providing **equal access** to expert instruction and **specialized courses** previously unavailable in rural areas
- Utilizes **efficient bandwidth optimization** to ensure reliable connectivity even with **low-speed internet**, making **education accessible** even during peak hours
- Improves access to **quality higher education** and **expert faculty** regardless of geographical location
- **Decreases financial burden** on rural institutions by **eliminating travel costs** for students and **reducing infrastructure requirements** while maintaining educational quality



- - **The Digital Divide and Its Impact on Education in Rural Areas** - Gandhi, G., & Umair, R.: This research examines how limited access to technology in rural regions hampers educational outcomes and explores challenges faced by students and teachers in rural areas due to digital infrastructure gaps. International Journal of Research Publication and Reviews, 2024
- - **Optimizing WebRTC Performance on Slow Networks** - WebRTC Ventures: This resource provides comprehensive strategies for codec selection, bandwidth management, and network optimization to enhance WebRTC applications under slow network conditions, crucial for rural connectivity scenarios.
- - **Do Virtual Schools Deliver in Rural Areas? A Longitudinal Study** Authors: D. Hamlin et al. Published In: Computers & Education, 2023: This longitudinal study demonstrates that rural students may have the most to gain from virtual schools and are more likely than urban/suburban students to enroll in virtual learning programs.